

General Mathematics - TERM 1

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taken from my notes in Obsidian; revamped.

Made with love and L^AT_EX

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“A joy in mathematics is being able to want more after one problem.”

Made possible by people like you.

Please read page 4 before proceeding.
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Preface

In my ninth year of junior high school, I gave away, every quarter, a reviewer of the mathematics that we learned every quarterly test. There were practice problems, a load of in-depth explanations, and frankly, that caused these guides to be pretty *long*. But my friends really appreciated these reviewers, because they had trouble understanding the material.

It has been a whole year since I have actually continued doing these notes. During the tenth year, I did not formulate a new set of reviewers because: a) they were time intensive. One reviewer would sacrifice a month and most of my time studying for next day's test, b) my mental health worsened ever since the end of the ninth year; I have major depressive episodes and coupling concerns of death, and c) I had no motivation.

Recently, ever since I enrolled under National University in East Ortigas, I have realized that this would be a great opportunity. Primarily because the material was slightly difficult this year. I also noticed that, with all due respect, a lot of people in this class struggle with mathematics.

I pride myself in joy teaching people mathematics, even if I am rusty and a little too confounding at times. But as someone who has done math as a hobby for years, I genuinely want to educate people.

So I would love to take this opportunity to give everyone a chance to be cleared up on confusions, understand more clearly, and understand things more firmly without stacking elliptic rocks that can quickly topple.

I firmly believe that understanding mathematics is always subjective. People understand faster without any additional material, and others need guidance to grasp. My goal isn't to find a solution or unite, it is so that it is easier for anyone to understand the topic. I want to make sure that it is cohesive but enough for anyone. Hopefully this has been sought after.

Before Proceeding (Important)

PLEASE KNOW: none of this document is spoken in Filipino. There will be a clear language barrier trying to understand what is being said, but I always attempt to use clear language that everyone can understand. Hopefully for the best.

PREREQUISITES: A CLEAR understanding of high school algebra, like factoring, as well as knowledge about integer arithmetic. Knowledge of Grade 10 mathematics, and simple conversions between two quantities. A quick refresher can be found on *my website*, specifically *here*. Use other resources too; see Appendix C (to be added later on).

Navigating The PDF

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There are also bookmarks by default. Click on the bookmark button in your reader and go to your page of choice.

I encourage you to also attempt the exercises, and all the gray boxes that are spread throughout the PDF to reinforce your understanding.

¹Most PDF readers can render these red boxes. If not, the contents are still clickable, or not dependent on your reader.

§1 Mathematics in Nature

In mathematics, patterns are found mostly everywhere. Most relate mathematics with numbers, but in general, nature is also related to mathematics.



Figure 1.1: The nautilus shell iconically references a logarithmic spiral, while the Windows 7 sample image of a chrysanthemum has an almost symmetric way of arranging petals.

§1.1 Patterns and Fibonacci

Formally, a **pattern** is a term to describe repeating objects, or events. Patterns in nature can be formed in two ways:

1. **Invoke-organized patterns** are patterns formed *with* external intervention.
2. **Self-organized patterns** are patterns formed *without* external intervention.

Patterns are not only just aesthetically pleasing, they also play an important role.

In 1202, a book would be published by Leonardo Pisano Bigollo Bonacci, titled *Liber Abaci* (Book of Calculations), he coined the **Fibonacci Sequence**. One of the most iconic sequences in all of mathematics, it derives from a simple thought Fibonacci had in his book:

Suppose a newly-born pair of rabbits (one male, one female) are put in a field. Rabbits are able to mate at the age of one month, so that at the end of the second month, a female can produce another pair of rabbits. Suppose that the rabbits never die, and that the female always produces one new pair (one male, one female) every month from the second month onward. How many pairs will there be in a year?

Every month, the pairs in total of that month, if laid in a sequence, create:

$$1, 1, 2, 3, 5, 8, \dots$$

On the first month, they can't breed yet, so there is still one pair. Same for the second month. But on the third, they make two new pairs. On the fourth, they make three. On the fifth, five. And so on.

These numbers make the **Fibonacci sequence**, a sequence of numbers wherein the next term is found by adding the previous term by the current. The first ten terms in the sequence are:

$$1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \dots$$

The reason why there is two 1's is because if there is no previous term from just 1, the sequence wouldn't form.

One of the most famous things the Fibonacci sequence is known for is the **Golden ratio**, which is taking the ratio of two terms in the sequence; term to the previous term.

EXPERIMENT - Find how approximate of the ratio is by taking the ratio of the 25th term to the 24th of the Fibonacci sequence is.

INSIGHT - If the numbers in the sequence get larger and larger, and take the ratios of any term and its succeeding term keep giving closer approximations to the golden ratio, how are both ideas correlated?

§1.2 Generating a Fibonacci Term

Definition 1.1: Formula for the n th Term of the Fibonacci Sequence

Given n is the place of the term:

$$F_n = \frac{(1 + \sqrt{5})^n - (1 - \sqrt{5})^n}{2^n \sqrt{5}}$$

Wherein F_n is the term of the sequence found.

§1.2.1 Examples

Example 1.1: Finding the n th term

Find the 50th term of the sequence.

SOLUTIONS

Simply note that our n is 50, and just substitute n :

$$F_n = \frac{(1 + \sqrt{5})^n - (1 - \sqrt{5})^n}{2^n \sqrt{5}}$$

$$F_{50} = \frac{(1 + \sqrt{5})^{50} - (1 - \sqrt{5})^{50}}{2^{50} \sqrt{5}}$$

Using a scientific calculator,

$$F_{50} = 1.258626902 \times 10^{10}$$

or 12586269020

Do not forget to check if your syntax is correct. Sometimes people would accidentally type the parentheses in the square root, causing an error upon computation.

§1.2.2 Problems

Given:

$$F_n = \frac{(1 + \sqrt{5})^n - (1 - \sqrt{5})^n}{2^n \sqrt{5}}$$

1. Find the 22nd term of the Fibonacci sequence.
2. Find the 123rd term of the sequence.
3. Find the 20th term of the sequence.
4. Find the 95th term of the sequence.
5. Find the average of the 20th term, and the 25th. *
6. Find the sum of the 67th term to the 69th.

Everything marked with an asterisk has solutions at the end of this PDF.

§1.3 A Brief Recall on Sequences

A **sequence** is a succession of numbers (**terms**) that follow a given pattern.

$$0, 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, \dots \quad (1)$$

This pattern is the list of all perfect squares.

A sequence can either be finite or infinite. An **infinite** sequence has no end, while **finite** means it has an end.

A sequence can be defined by two forms: A sequence with an **explicit formula** means the next term is unique; it doesn't involve a previous term, as it just succeeds the sequence normally. They follow a general rule, that rule being an equation.

For example, this sequence:

$$7, 12, 17, 22, 27, \dots$$

Follows the simple pattern $a_n = 5n + 2$, wherein n is your given term of choice. If I were to find the 10th term in that sequence, $a_{10} = 5(10) + 2$, it would be 52.

A sequence with a **recurrence relation** is when the sequence uses previous terms to make the next one. For example, the Fibonacci sequence is considered a recurrence relation because you find the next term by adding the last two terms in the sequence.

Recurrence relations have an inductive definition to the formula for the sequence. You can use any letter for each sequence, but for simplicity, I will use u_n for recurrence and a_n for explicit.

§1.3.1 Examples

Example 1.2: Finding a List of Terms

Given the formula of $a_n = 6n - 1$, find the first 5 terms of the sequence.

SOLUTIONS

The most simplest way would be to create a table:

Term Number	Formula	Result
a_1	$a_1 = 6(1) - 1$	5
a_2	$a_2 = 6(2) - 1$	11
a_3	$a_3 = 6(3) - 1$	17
a_4	$a_4 = 6(4) - 1$	23
a_5	$a_5 = 6(5) - 1$	29

Therefore, the first five terms of the sequence would be $\boxed{5, 11, 17, 23, 29}$.

Example 1.3: Recurrence Relation

Given the recurrence relation $u_n = 2u_{n-2} + 3u_{n-1}$, find the next three terms, given that u_1 and u_2 are both equal to 1.

SOLUTIONS

Term Number	Formula	Result
u_3	$u_3 = 2(1) + 3(1)$	5
u_4	$u_4 = 2(1) + 3(5)$	16
u_5	$u_5 = 2(5) + 3(16)$	58

Therefore, the next three terms after 1, 1 would be $\boxed{5, 16, 58}$.

EXPERIMENT - Make your own explicit formula and recurrent relation. If making a recurrent relation, do not forget to include a given first term, or second term.

INSIGHT - Compare $a_n = 4n + 5$ to $u_n = 4u_{n-1} + 5$. Which one grows faster?

§1.3.2 Problems

I - Enumeration

List down the first five terms, and the next five terms of the following explicit formulas and recurrence relations respectively.

1. $a_n = 10n - 13$

2. $a_n = n^2 - 1$

3. $a_n = \frac{1}{n}$

4. $a_n = \frac{n}{n+1}^*$

5. $a_n = n^2 - n - 1$

6. $a_n = n^3 - n^*$

7. $u_n = u_{n-1} + 2$
given $u_1 = 12$.

8. $u_n = 2u_{n-1}^2 - u_{n-2}$
given $u_1 = 2, u_2 = 8$.*

9. $u_n = \frac{u_{n-1}}{2} + 3$
given $u_1 = -6$.

10. $u_n = 1 - \frac{1}{2}u_{n-1}^2$
given $u_1 = 1$. *

11. $u_n = u_{n-2}(u_{n-1} + 3)$
given $u_1 = -5$ and $u_2 = 0$.

12. $u_n = u_{n-1} + u_{n-2}$
given $u_1 = 1$ and $u_2 = 1$.

II - Word Problems

Provide a table of all of the following problems.

- Clark had P231.5 in his piggy bank. He fills it up with thrice that amount every six months. List down how much his piggy bank was every six months in the last two years.*
- An excel sheet embedded in a website documents the list of all applicants to an event starting from 250 already pre-signed. The first row had 25 applicants. The next there was 45. Then 65. List down the next five rows of applicants.
- Rin plays an online game with a kill counter for the session. Her matches end every thirty minutes. Her kill counter was at 5 when she won her first game. The next game she won 11. After that, 17. List down how many kills she got in the next two and a half hours.
Hint: $a_n = 6n - 1$

4. Gura created a program that can do a task at an increasing rate of efficiency every day. She started her program with a rate of 128 processes a day. The number doubles every day, so after 5 days, list down the speeds the program has gotten.
5. A vineyard maintainer hones in 180 grapes every harvest. By the first harvest, there would be 189 grapes. The next would be 369. What would be the maintainer's list of grapes collected per harvest, in the next five harvests?
6. A generative model, created by a hobbyist, is tasked to scrape the web for information "ethically." Initially it was slow, at around 2000 lines every ten minutes. However, every ten minutes, the speed would double and increase by five hundred. How fast would the model be after an hour?

§1.4 Finding the Explicit Formula for a Sequence

A sequence's formula can be found by either induction or through mathematical means.

Inductive reasoning involves the use of patterns to create a conclusion. For example,

$$1, 2, 2, 3, 3, 3, 4, 4, ?, ?, \dots$$

The pattern starts with "one 1," then "two 2's," then "three 3's." By induction, the missing terms would be both 4, because it would be "four 4's."

However, inductive reasoning sometimes does not always yield quick and surefire results.

That's why there is a method for finding a sequence's rule.

Methodically, this method is closely related to finding a quadratic equation through a list of outputs. For example,

$f(x) = x^2 + x$	Output
1	2
2	6
3	12
4	20

We can rewrite the outputs as a sequence of numbers.

$$2, 6, 12, 20, \dots$$

Given this information, what if we didn't know what the equation was? Let's say we were given just this list of numbers.

Firstly, we should create a table with the first three terms in succession.

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Term	2	6	12

On the next row, we take the difference of the second term to the first, and the third to the second.

Term	2	6	12
First difference	4	6	

Then we take the difference of the two differences that we got.

Term	2	4	12
First difference	4	6	
Second difference	2		

Closely follow: Equate the third term to $2a$, equate the second term to $3a + b$, and equate the first term to $a + b + c$. All-in-all, we get a linear system wherein we solve for each variable.

$$\begin{cases} 2a = 2 \\ 3a + b = 4 \\ a + b + c = 2 \end{cases}$$

All we have to do is solve each one-by-one. In this method, the process is always the same.

1. Solve for a .
2. Substitute the result for a into $3a + b$ and solve for b .
3. Substitute the results of both a and b into $a + b + c$ and solve for c .

SOLUTIONS

We follow these steps.

1. Solve for a .

$$2a = 2$$

$$a = 1$$

2. Substitute the result for a into $3a + b$ and solve for b . Given $a = 1$:

$$3(1) + b = 4$$

$$b + 3 = 4$$

$$b = 1$$

3. Substitute the results of both a and b into $a + b + c$ and solve for c .
Given we now know $a = 1$ and $b = 1$:

$$1 + 1 + c = 2$$

$$c + 2 = 2$$

$$c = 0$$

Now we know that $\boxed{a = 1, b = 1, c = 0}$. The reason why we find these numbers is because we are trying to satisfy the general equation for a quadratic $ax^2 + bx + c$ (x can be any variable.) Given this information, we can substitute our a, b, c to the equation.

$$\begin{aligned} ax^2 + bx + c &= (1)x^2 + (1)x + 0 \\ &= \boxed{x^2 + x} \end{aligned}$$

This is the original equation I used as an example.

Basically, this method helps us identify any equation of a sequence. It can find both linear and quadratic expressions, also given that it is an equation that yields the same list of results.²

²Any equation can have the same result; $1 + 1 = 2$ and $-9 + 11 = 2$. A cubic equation (polynomial degree 3) can have the same results as a regular quadratic.

§1.4.1 Examples

Example 1.4

Find the equation for the sequence $0, 12, 42, \dots$

Remember, the method remains the same. $0, 12, 42, \dots$

Term	0	12	42
First difference	12	30	
Second difference	18		

$$\begin{cases} 2a = 18 \\ 3a + b = 12 \\ a + b + c = 0 \end{cases}$$

1. $2a = 18$

$$\begin{aligned} 2a &= 18 \\ a &= 9 \end{aligned}$$

2. $3a + b = 12$ given $a = 9$

$$\begin{aligned} 3(9) + b &= 12 \\ b + 27 &= 12 \\ b &= -15 \end{aligned}$$

3. $a + b + c$ give $a = 9, b = -15$

$$\begin{aligned} 9 + (-15) + c &= 0 \\ 9 - 15 + c &= 0 \\ c - 6 &= 0 \\ c &= 6 \end{aligned}$$

4. Substitute.

$$an^2 + bn + c = (9)n^2 + (-15)n + 6$$
$$9n^2 - 15n - 3 = \boxed{3(3n^2 - 5n + 2)}$$

5. Verify:

$$a_1 = 3(3(1)^2 - 5(1) + 2) = 0✓$$

$$a_2 = 3(3(2)^2 - 5(2) + 2) = 12✓$$

$$a_3 = 3(3(3)^2 - 5(3) + 2) = 42✓$$

In Simple Terms:

1. List down the first three terms of a sequence.
2. Take the difference of the second term to the first, and the third term to the second.
3. Take the difference of your differences.
4. Equate $2a$ to your second difference, equate $3a + b$ to difference between the second term to the first, and equate $a + b + c$ to the first term of the sequence.
5. Solve for a, b, c .
6. Substitute your values for a, b, c to $an^2 + bn + c$ to find your equation.
7. Verify to check if you did anything wrong in your solution.

§1.4.2 Problems

Find the explicit formula of the following sequences:

1. $11, 16, 23, \dots$

9. $\frac{5}{2}, \frac{11}{2}, \frac{21}{2}, \dots$ *

2. $8, 17, 28, \dots$

10. $1, 0, 1, \dots$

3. $-2, 5, 18, \dots$

11. $\frac{7}{4}, \frac{31}{4}, \frac{71}{4}$

4. $1, 13, 37, \dots$

12. $8, 13, 18, \dots$

5. $\frac{1}{2}, 2, \frac{9}{2}, \dots$

13. $10, 21, 32, \dots$

6. $0, 3, 8, \dots$

14. $-9, -7, -5, \dots$

7. $0, 1, 4, \dots$

15. $2, 5, 9, \dots$ *

8. $0, \frac{3}{2}, 4, \dots$ *

16. $23, 15, 3, \dots$

§1.5 The Series

A **series** is the sum of a sequence. it is represented as S_n .

A series is **infinite**, S_∞ , is a series with no end. A **finite** series, represented as S_n , has an end.

Formally, it is represented through summation notation, with the symbol being capital sigma.

Definition 1.2: Summation Notation

$$\sum_{i=n}^j a_i$$

wherein i is the index, n is the first value of i , j is the last value of i , and a_i is the function or rule that the sum follows.

i is just a *dummy variable*, meaning it just means can be replaced by any variable.

Remark.

Infinite series can have sums, but not all of them do. An infinite series is *convergent* if it has a sum, *divergent* if not.

§1.5.1 Examples

Example 1.5: Finding the Sum

Find the following sum.

$$\sum_{x=3}^6 x^2 + 1$$

The terms in the series are obtained from substituting all numbers from i to j , so in this instance, we substitute all of x with numbers from 3 to 6 in a_x .

SOLUTIONS

$$\sum_{x=3}^6 x^2 + 1 = ((3)^2 + 1) + ((4)^2 + 1) + ((5)^2 + 1) + ((6)^2 + 1)$$

Evaluate.

$$(9 + 1) + (16 + 1) + (25 + 1) + (36 + 1) = 10 + 17 + 26 + 37 \\ = \boxed{90}$$

Explanation (What?).

The summation notation works by taking the explicit formula of a sequence, recall is a rule that follows an equation, which in this case was $a_n = n^2 + 1$ (I just replaced the x with an n), and added a certain part of the sequence's terms. If I were to list this sequence starting from the first term,

$$2, 5, 10, 17, 26, 37, 50, \dots$$

All I did was to add all terms from the third term to the sixth. That's how the summation notation works. All it does is to sum a certain part of a sequence given the explicit formula.

Example 1.6: Combining Sums

Evaluate the following.

$$\left(\sum_{x=1}^5 x - 2 \right) - \left(\sum_{x=1}^5 1 - 3x \right)$$

A simple property of summations is that adding two sums with the same start and final numbers but different explicit formulas, we can consider the sum to be:

$$\left(\sum_{i=n}^j a_i\right) \pm \left(\sum_{i=n}^j b_i\right) = \sum_{i=n}^j a_i \pm b_i$$

Combine both summations if their starting and ending numbers granted are the same. Given this information, we should also remember that, if the sign is *negative*, do not forget to distribute the negative sign in the equation.

SOLUTIONS

$$\begin{aligned} \left(\sum_{x=1}^5 x - 2\right) - \left(\sum_{x=1}^5 1 - 3x\right) &= \sum_{x=1}^5 x - 2 - (1 - 3x) \\ \sum_{x=1}^5 x - 2 - 1 + 3x &= \sum_{x=1}^5 4x - 3 \\ (4(1) - 3) + (4(2) - 3) + (4(3) - 3) + (4(4) - 3) + (4(5) - 3) \\ 1 + 5 + 9 + 13 + 17 &= \boxed{45} \end{aligned}$$

§1.5.2 Problems**I - Simple Sums**

Evaluate all the following sums.

1. $\sum_{x=2}^7 x$

4. $\sum_{e=6}^8 e^2 + e$

7. $\sum_{n=10}^{13} \frac{n}{n-1}$

2. $\sum_{x=3}^8 x + 120$

5. $\sum_{p=7}^{10} \frac{p+1}{p}$

8. $\sum_{q=-1}^4 \frac{2-q}{q+2}^*$

3. $\sum_{k=1}^3 k + \frac{1}{k}$

6. $\sum_{x=4}^8 \frac{x^2}{2}$

9. $\sum_{b=3}^6 \frac{b^2}{(-1)^b}^*$

II - Sums of Sums

Evaluate.

1. $\left(\sum_{r=1}^3 r - 1 \right) + \left(\sum_{r=1}^3 2r + 3 \right)$

2. $\left(\sum_{x=-2}^2 x^2 - x \right) + \left(\sum_{x=-2}^2 2x - x^2 \right)$

3. $\left(\sum_{x=3}^7 x^2 + 2 \right) - \left(\sum_{x=3}^7 x(x-1) \right)^*$

4. $\left(\sum_{k=1}^5 \frac{k}{2} \right) + \left(\sum_{k=1}^5 k \right)^*$

5. $\left(\sum_{u=-1}^3 -u + \frac{1}{2} \right) - \left(\sum_{u=-1}^3 -u^2 - u \right)$

§1.6 Arithmetic and Geometric Sequences Revisited

In Grade 10, this topic was the very first to be discussed of the whole school year. My class of twelve found it quite simple, but if it isn't for you, don't worry. It is normal.

An **arithmetic sequence** is a sequence that has a pattern with a **common difference**. The common difference is the universal result of subtracting a term by its previous. Meaning the sequence has the next term succeed by a certain number.

$$3, 7, 11, 15, 19, \dots$$

This arithmetic sequence has a common difference of 4.

A **geometric sequence** is a sequence that has a pattern of a **common ratio**. A common ratio has the same idea as a common difference, but instead we divide a term by its previous.

$$4, 12, 36, 108, 324$$

This geometric sequence has a common ratio is 3.

We shall begin with the first theorem.

Theorem 1.1: Finding the n th Term in an Arithmetic Sequence

Given a_n being the n th term, a_1 being the first term, and d being the common difference, to find the n th term of an arithmetic sequence is given as:

$$a_n = a_1 + (n - 1)d$$

n is the place of your term, for example if $n = 4$, then you are at the fourth term.

For a geometric sequence,

Theorem 1.2: Finding the n th Term in a Geometric Sequence

Given a_n being the n th term, a_1 being the first term, and r being the common ratio, to find the n th term of an arithmetic sequence is given as:

$$a_n = a_1 r^{n-1}$$

§1.6.1 Examples

Example 1.7: Arithmetic Sequence

Find the following:

1. The 7th term of 9, 15, 21, 27, ...
2. The 10th term of $-1, -6, -11, -16, \dots$
3. The 15th term of 2, 17, 32, 47, ...

SOLUTIONS

1. Given $a_1 = 9$, $n = 7$, and the common difference d is 6,

$$a_n = a_1 + (n - 1)d$$

$$a_7 = 9 + (7 - 1)(6)$$

$$a_7 = 9 + (6)(6)$$

$$a_7 = \boxed{45}$$

The 7th term of the sequence is 45.

2. Given $a_1 = -1$, $n = 10$, and the common difference is -5 (because $-6 - (-1) = -5$,

$$\begin{aligned}a_n &= a_1 + (n - 1)d \\a_{10} &= -1 + (10 - 1)(-5) \\a_{10} &= -1 + (9)(-5) \\a_{10} &= -1 - 45 \\a_{10} &= \boxed{-46}\end{aligned}$$

The 10th term in the sequence is -46 .

3. Given $a_1 = 2$, $n = 15$, and the common difference is 15 ,

$$\begin{aligned}a_n &= a_1 + (n - 1)d \\a_{15} &= 2 + (15 - 1)(15) \\a_{15} &= 2 + (14)(15) \\a_{15} &= \boxed{210}\end{aligned}$$

The 15th term of the sequence is 210 .

Example 1.8: Geometric Sequence

Find the following:

1. The 10th term of $4, 8, 16, 32, \dots$
2. The 17th term of $5, 30, 180, 1080, \dots$
3. The 21st term of $18, 6, 2, \frac{2}{3}, \dots$

SOLUTIONS

1. Given that $a_1 = 4$, $n = 10$, and the common ratio r being 2 :

$$\begin{aligned}a_n &= a_1 r^{n-1} \\a_{10} &= (4)(2^{10-1}) \\a_{10} &= 4(2^9) \\a_{10} &= \boxed{2048}\end{aligned}$$

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The tenth term in the sequence is 2048.

2. Given that $a_1 = 5$, $n = 17$, and the common ratio being 6:

$$a_n = a_1 r^{n-1}$$

$$a_{17} = (5)(6^{17-1})$$

$$a_1 7 = 5(6^{16})$$

$$a_1 7 = \boxed{14105549537280}$$

The seventeenth term in the sequence is 14,106,549,537,280.

3. Given that $a_1 = 18$, $n = 21$, and the common ratio being $\frac{1}{3}$:

$$a_n = a_1 r^{n-1}$$

$$a_{21} = (18) \left(\frac{1}{3} \right)^{21-1}$$

$$a_{21} = (18) \left(\frac{1}{3} \right)^{20}$$

$$a_{21} = 18 \cdot \frac{1}{3486784401}$$

$$a_{21} = \frac{18}{3486784401}$$

$$a_2 1 = \boxed{\frac{6}{1162261467}}$$

The twenty-first term of the sequence is $\frac{6}{1162261467}$.

Example 1.9: An Interesting Example

This was an example our lecturer gave.

In an arithmetic sequence, the fourth term is 12, and the twentieth is 92. Given this information only, what is the 25th term?

The information given is not complete. We don't have a common difference, nor do we have the first term. To find these terms, we can write each statement with their incomplete parts: given we have the n th terms 12 and 92, and n which is 4 and 20 respectively,

$$12 = a_1 + (4 - 1)d \text{ and } 92 = a_1 + (20 - 1)d$$

Simplify,

$$12 = a_1 + 3d \text{ and } 92 = a_1 + 19d$$

Given we are solving for both a_1 and d , what we have is a linear system of equations.

$$\begin{cases} a_1 + 3d = 12 \\ a_1 + 19d = 92 \end{cases}$$

There are two ways we can solve this: either by a) elimination or b) substitution. What I will do is through substitution. I suggest using elimination for a much more quicker solution.

SOLUTIONS

First, solve for either a_1 or d on either equation; I will use a_1 on the first equation:

$$\begin{aligned} a_1 + 3d &= 12 \\ a_1 + \cancel{3d} - \cancel{3d} &= 12 - 3d \\ a_1 &= 12 - 3d \end{aligned}$$

General Mathematics

Substitute $a_1 = 12 - 3d$ in the second sequence. Solve for d , because if we substitute a_1 to the equation, there will only be d as the main variable.

$$\begin{aligned}a_1 + 19d &= 92 \\(12 - 3d) + 19d &= 92 \\12 - 3d + 19d &= 92 \\12 + 16d &= 92 \\16d &= 80 \\d &= 5\end{aligned}$$

Now we have our common difference of 5. Substitute this to either equation; I will substitute this to the first.

$$\begin{aligned}a_1 + 3d &= 12 \\a_1 + 3(5) &= 12 \\a_1 + 15 &= 12 \\a_1 &= 12 - 15 \\a_1 &= -3\end{aligned}$$

Now we have $\boxed{a_1 = -3}$ given $\boxed{d = 5}$. All we have to do now is find the twenty-fifth term of the sequence.

$$\begin{aligned}a_n &= a_1 + (n - 1)d \\a_{25} &= (-3) + (25 - 1)(5) \\a_{25} &= -3 + 5(26) \\a_{25} &= \boxed{127}\end{aligned}$$

Therefore, the twenty-fifth term of the sequence is 127.

§1.6.2 Problems

I - Arithmetic Sequences

Solve for what is given.

1. The 12th term of $-4, 12, 28, 44, \dots$
2. The 9th term of $-7, -21, 35, 49, \dots$
3. The 18th term of $14, -11, -36, -61, \dots$
4. The 23rd term of $\frac{3}{2}, 4, \frac{13}{2}, 9, \dots$
5. The 40th term of $-360, -353, -346, -339, \dots$
6. Find the 100th term given the common difference is -3 , the first term is 2.
7. Find the 16th term given the common difference is 9, and the first term is the twentieth multiple of 3.
8. Find the 35th term given the common difference is the fifth prime number, and the first term is its 7th multiple.*
9. Find the common difference if given $a_1 = 2$, $n = 11$, and $a_n = 72$. *Hint: solve for d*
10. Find n if the first term is 20, the common difference is 12, and $a_n = 212$.*

II - Geometric Sequences

Solve for what is given.

1. The 8th term of $-3, 9, -27, 81, \dots$
2. The 8th term of $5, 15, 45, 135, \dots$
3. The 14th term of $\frac{13}{3}, \frac{26}{3}, \frac{52}{3}, \frac{104}{3}, \dots$ *
4. The 16th term of $10, 90, 8100, 7290, \dots$
5. The 33rd term of $\frac{2}{5}, 2, 10, 50, \dots$
6. The sequence of the powers of 2 is 2, 4, 8, 16, 32, $\dots$. Find the 16th power of two.

7. Find the 10th term if the first term is 4 and the common ratio is 11.
8. Find the 15th term if the first term is 40 and the common ratio is 6.
9. Find the first term given $a - n = 960$, $n = 4$, and the common ratio r is also 4.
10. Find the common ratio, given $a_1 = 4$, and that the eighth term is 312500.

III - Word Problems

Solve for what is indicated in each problem.

1. Imagine in a hypothetical medium, 20 bacteria can reproduce at a rate of 14 per ten minutes. After 60 minutes, how much bacteria will there be?
2. My Chemical Romance will play the Philippine Arena at November 14, 2026. The total capacity of the arena is 55,000, however it was reduced to accommodate the stage set for the show. Every hour, after an initial 7520 enter for VIP, 8000 concert-goers go to the arena. In 6 hours, the venue be full. How many slots were reduced for the show?*
3. 960 pieces of paper were cut in total. There were a certain amount of papers that Mei bought as an experiment. She cut each paper into four pieces, and into four pieces again. She did this two more times. How many papers were there initially?
4. Fifty thousand and five hundred meals will have been donated to victims of a typhoon. The first day, 2500 meals were donated. 1600 more meals will be donated additionally from the day before due to volunteers of the local community helping. How many days will it take to consume all four-hundred thousand meals?

BONUS - Hard Question

This is not required.

A music festival will handle thirty thousand participants, including the bands. 1000 people who are in bands arrive early to prepare their sets. Later, 875 people arrive.

Every hour, the venue's attendees quadruple how many arrived the previous hour. *How many hours does it take for the venue to be fully packed?*

§1.7 Arithmetic and Geometric Series

Recall that a series is a sum of a sequence. We will apply this to arithmetic and geometric sequences.

An **arithmetic series** is the sum of an n amount of terms in an arithmetic sequence. A **geometric series** is the sum of an n amount of terms in a geometric series.

What was taught in class does not involve the summation notation, but we will use it anyway for formality.

Theorem 1.3: Sum of First n Terms in an Arithmetic Sequence

Given this sum,

$$\sum_{n=1}^m a_1 + (n-1)d$$

For any n terms, given a_1 , and a_n , the sum of first n terms S_n is:

$$S_n = n \left(\frac{a_1 + a_n}{2} \right)$$

If a_n is not given, but d is also given, then:

$$S_n = \frac{n}{2}(2a_1 + (n-1)d)$$

Sometimes, when asked, there is no final term a_n for the sum to add up to given. We use the second formula if so. However, if the final term is given, one may choose either formula, but is much more easier to use the first.

Geometric series have the same idea. However, geometric series can also be infinite, unlike an arithmetic series. This will be explained later.

Theorem 1.4: Sum of First n Terms in a Geometric Sequence

Given this sum,

$$\sum_{n=1}^m a_1 r^{n-1}$$

For any n terms, given a_1 , r , and n , the sum of first n terms S_n is:

$$S_n = a_1 \left(\frac{1 - r^n}{1 - r} \right)$$

An **infinite geometric series** is the sum of an infinite geometric sequence. Yes. The ENTIRE thing. However, there are limitations.

Earlier, in a Remark I made, I said that infinite series *can* have sums. Just not all of them. An infinite series **diverges** if it does not have a sum. However, if it does, it **converges** given that the absolute value of the common ratio is less than 1 $|r| < 1$. There are sums that do have a common ratio greater than 1, yet converge, however those are tackled in courses related to calculus or higher mathematics, so those will be ignored for now.

Theorem 1.5: The Sum of an Infinite Geometric Series

Given this sum,

$$\sum_{n=1}^{\infty} a_1 r^{n-1}$$

If $|r| < 1$,

$$S_{\infty} = \frac{a_1}{1 - r}$$

§1.7.1 Examples

Example 1.10: Arithmetic Series

Find the following sums:

1. $\sum_{n=1}^{10} 2n - 5$
2. Given $a_n = 20$, and $a_1 = 1$, find the sum of the first 40 terms in the sequence.

SOLUTIONS

1. Don't let the first equation scare you. We can simply enumerate a sequence by listing down the terms created. It's just asking the sum of the first ten terms given the rule.

$$\begin{aligned}a_n &= 2n - 5 \\ &= -3, -1, 1, 3, \dots\end{aligned}$$

Now we know that $a_1 = -3$, $d = 2$, and $n = 10$. We can sum this by hand, but let's use the formula instead. There is no a_n given, so we resort to the second formula.

$$\begin{aligned}S_n &= \frac{n}{2}(2a_1 + (n-1)d) \\ S_{10} &= \frac{10}{2}(2(-3) + (10-1)(2)) \\ S_{10} &= 5 \cdot (-6 + 18) \\ S_{10} &= 5 \cdot 12 \\ S_{10} &= \boxed{420}\end{aligned}$$

The sum of the first ten terms of the given series is 420. Nice.

2. Use the first formula.

$$S_n = n \left(\frac{a_1 + a_n}{2} \right)$$

$$S_{40} = 40 \cdot \left(\frac{1 + 20}{2} \right)$$

$$S_{40} = 40 \cdot \frac{21}{2}$$

$$S_{40} = \frac{840}{2}$$

$$S_{40} = \boxed{420}$$

Wow, both of them are 420. I made these up on the fly.

Example 1.11: Geometric Series

Find the following sums:

1. Find the sum of the first 15 terms in 4, 20, 100, 500, ...

2. $\sum_{n=1}^{\infty} \frac{1}{(-2)^n}$

SOLUTIONS

1. Use the formula.

$$S_n = a_1 \left(\frac{1 - r^n}{1 - r} \right)$$

$$S_{15} = 4 \cdot \frac{1 - 5^{15}}{1 - 5}$$

$$S_{15} = 4 \cdot \frac{\cancel{30517578124}}{\cancel{4}}$$

$$S_{15} = \boxed{7629394531}$$

2. The sum goes to infinity, so we have to use the formula for an infinite geometric series' sum. List down the terms first.

$$-\frac{1}{2}, \frac{1}{4}, -\frac{1}{8}, \frac{1}{16}, \dots$$

You can tell that the sum alternates between positive and negative. This is a special kind of series that applies to both kinds, called an *alternating series*. To get the common ratio, the absolute value of $-\frac{1}{2}$ (the initial ratio) is $\frac{1}{2}$ or just switching the negative to positive. Remember, the sum for an infinite geometric series is only applicable if $|r| < 1$.

$$S_{\infty} = \frac{a_1}{1 - r}$$

$$S_{\infty} = \frac{-\frac{1}{2}}{1 - \frac{1}{2}}$$

$$S_{\infty} = \frac{-\frac{1}{2}}{\frac{1}{2}}$$

$$S_{\infty} = \boxed{1}$$

EXERCISE - What is the sum of $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} \dots$?

INSIGHT - Why does the sum in the previous question equate to its answer?

§1.7.2 Problems

I - Arithmetic Series

Solve for the indicated. If there is no a_n , use the second formula. If it is expressed in summation form, express the first 5 terms first before solving.

1. $a_n = 90$, $n = 10$, and $a_1 = -18$. Find the total sum.
2. $a_n = 200$, $d = 17$, $a_1 = 13$, $n = 12$. Find the total sum.
3. $d = 6.5$, $a_1 = 16.5$, $n = 16$. Find the total sum.*
4. The ninth term in a sequence is 45, while the first is 101. Find the sum of all terms within that range.
5. $a_n = -59$, $a_1 = -2$, $n = 20$. Find the total sum.
6. $a_n = 135$, $a_1 = 15$, $n = 27$. Find the total sum.
7. Given $-14, -20, -26, -32, \dots$, find the sum of the first 30 terms.
8. $\sum_{n=1}^{20} 3(n+5)$
9. $\sum_{n=1}^9 \frac{n+1}{2}$
10. $\sum_{n=1}^{29} n+3$

II - Geometric Series

Solve for the indicated. If it is expressed in summation form, write down the first three terms before solving.

1. 10, 100, 1000, $\dots$. Find the sum of the first ten terms.
2. 20, 10, 5, $\frac{5}{2}$. Find the sum of the first thirteen terms.
3. $\frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots$. Find the sum.
4. 16, 8, 4, 2, 1, $\dots$. Find the sum of the first twenty terms.
5. 5, -35, 245, -1715, $\dots$. Find the sum of the first six terms.
6. $\frac{1}{5} + \frac{1}{25} + \frac{1}{125} + \dots$. Find the sum.

7. $\sum_{n=1}^7 2(3^{n-4})$

9. $\sum_{n=1}^{\infty} \frac{2}{5^{k+2}}$

8. $\sum_{n=1}^8 \frac{1}{4^{n+4}} *$

10. $\sum_{n=1}^{\infty} \frac{1}{(-10)^n}$

III - Word Problems

Solve for indicated. Note there are no problems that involve the infinite series formula.

1. Yenny has 35 coins in her piggy bank. She adds 19 coins every week. By the 18th week, how many coins does she have?
2. A moderately sized pool was built in a canonically rich family's home. Their water system however, doesn't work properly. For some reason, it runs off magic and the pool fills up comically fast.

The water fills up $\frac{7}{16}$ feet of the pool once its on. In five seconds, it fills up $\frac{7}{8}$ feet. After another five seconds, $\frac{7}{4}$ feet. After 25 seconds, how much water (in terms of feet) would have been dispensed in total?

3. Garrett is a schizophrenic. He aims to fish a lot over the summer break. As in hundreds. He catches 10 fish the first week. He gathers 18 more, then 26 the next two weeks. After eight weeks, how much fish has garret caught? And was he delusional?

BONUS - Intermediate Question

The game 2048 is known for its addicting nature, and was popular a decade ago or so. In the digital age, a robot can be capable of playing this game. It starts slow with the robot getting to win a game after an hour, after 15 losses. The next hour, the robot wins two games with 30 losses. The next? Four games won after 45 losses. After running for half a day, how many games has the robot lost, how many won, and how many more games did the robot lose compared to their wins?

Appendices

§A Part One

§A.1 Integer Arithmetic

Multiplying/dividing opposite signs will always be negative.

$$(-) \cdot (+) = (-)$$

$$(+) \cdot (-) = (-)$$

Multiplying/dividing similar signs will always be positive.

$$(+) \cdot (+) = (+)$$

$$(-) \cdot (-) = (+)$$

Adding or subtracting negative numbers can be done as the following: a and b can be any number.

$$a + (-b) = a - b$$

$$-a + b = b - a$$

$$(-a) + (-b) = -(a + b)$$

For subtracting,

$$a - (-b) = a + b$$

$$-a - b = -(a + b)$$

$$-a - (-b) = b - a$$

If you ever see $a + (-b)$, it is always the same as $a - b$. That's why it is sometimes simplified to that.

Whatever sign has the bigger number from the two, will be the sign of the result.

§A.2 Solving Linear Systems

Given a linear system,

$$\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases}$$

wherein the coefficients are any arbitrary number, a system of linear equation can be solved by elimination or substitution. I will show the method of substitution.

1. Solve for x or y on either equation. I will pick x on the first equation.

$$\begin{aligned} a_1x + b_1y &= c_1 \\ a_1x &= c_1 - b_1y \\ x &= \frac{c_1 - b_1y}{a_1} \end{aligned}$$

2. Substitute $x = \frac{c_1 - b_1y}{a_1}$ to the second equation if you chose the first equation. Vice versa if the second. Solve for y if you chose to solve for x for step 1.

$$\begin{aligned}
 a_2x + b_2y &= c_2 \\
 a_2 \left(\frac{c_1 - b_1y}{a_1} \right) + b_2y &= c_2 \\
 \frac{a_2(c_1 - b_1y)}{a_1} + b_2y &= c_2 \\
 \frac{a_2(c_1 - b_1y)}{a_1} + \frac{b_2a_1}{a_1} &= c_2 \\
 \frac{a_2(c_1 - b_1y) + b_2a_1}{a_1} &= c_2 \\
 \frac{a_2c_1 - a_2b_1y + b_2a_1}{a_1} &= c_2 \\
 \frac{a_2c_1 - b_2a_1}{a_1} - \frac{a_2b_1y}{a_1} &= c_2 \\
 \frac{a_2c_1 - b_2a_1}{a_1} - c_2 &= \frac{a_2b_1y}{a_1} \\
 \cancel{a_1} \cdot \frac{a_2c_1 - b_2a_1 - c_2a_1}{\cancel{a_1}} &= \cancel{a_1} \cdot \frac{a_2b_1y}{\cancel{a_1}} \\
 a_2c_1 - b_2a_1 - c_2a_1 &= a_2b_1y \\
 y &= \frac{a_2c_1 - b_2a_1 - c_2a_1}{a_2b_1} \\
 y &= \frac{a_2c_1 - a_1(b_2 - c_2)}{a_2b_1}
 \end{aligned}$$

3. Once y is found, substitute y back into x .³

4. Now both x and y are found. Done.

³I am NOT writing all that out. It is genuinely so syntax intensive.

§A.3 Solutions to Selected Problems

§A.3.1 Sections 1.2.2 and 1.3.2

1.2.2

$$F_{20} = \frac{(1 + \sqrt{5})^{20} - (1 - \sqrt{5})^{20}}{2^{20}\sqrt{5}}$$

$$F_{20} = 6765$$

$$F_{25} = \frac{(1 + \sqrt{5})^{25} - (1 - \sqrt{5})^{25}}{2^{25}\sqrt{5}}$$

$$F_{25} = 75025$$

Find the average.

$$\frac{6765 + 75025}{2} = \frac{81790}{2} = \boxed{40895}$$

1.3.2

$$4. \quad \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \dots$$

$$6. \quad 0, 6, 24, 60, 120, \dots$$

$$8. \quad 0, 0, 0, 0, 0, 0, \dots$$

$$10. \quad \frac{1}{2}, \frac{7}{8}, \frac{79}{128}, \frac{13983}{16384}, \frac{72911167}{268135456}, \dots$$

	Half-year	Amount
1.	1	231.5
	2	463
	3	694.5
	4	926

§A.3.2 Sections 1.4.2 and 1.5.2

1.4.2

8.

Term	0	$\frac{3}{2}$	4
First difference	$\frac{3}{2}$	$\frac{5}{2}$	
Second difference	1		

$$\begin{cases} 2a = 1 \\ 3a + b = \frac{3}{2} \\ a + b + c = 0 \end{cases}$$

$$a = \boxed{\frac{1}{2}}$$

$$b = \boxed{0}$$

$$c = \boxed{-\frac{1}{2}}$$

$$= an^2 + bn + c$$

$$= \boxed{\frac{1}{2}n^2 - \frac{1}{2}}$$

$$\text{or } \frac{n^2 - 1}{2}$$

9.

Term	$\frac{5}{2}$	$\frac{11}{2}$	$\frac{21}{2}$
First difference	3	5	
Second difference	2		

$$\begin{cases} 2a = 2 \\ 3a + b = 3 \\ a + b + c = \frac{5}{2} \end{cases}$$

$$\begin{aligned}
 a &= \boxed{1} \\
 b &= \boxed{0} \\
 c &= \boxed{\frac{3}{2}} \\
 &= an^2 + bn + c \\
 &= \boxed{n^2 + \frac{3}{2}}
 \end{aligned}$$

15.

Term	2	5	9
First difference	3	4	
Second difference	1		

$$\begin{cases}
 2a = 1 \\
 3a + b = 3 \\
 a + b + c = 2
 \end{cases}$$

$$\begin{aligned}
 a &= \boxed{\frac{1}{2}} \\
 b &= \boxed{\frac{3}{2}} \\
 c &= \boxed{0} \\
 &= an^2 + bn + c \\
 &= \boxed{\frac{1}{2}n^2 + \frac{3}{2}x}
 \end{aligned}$$

1.5.2

Sums can be evaluated manually.

8.

$$\sum_{q=-1}^4 \frac{2-x}{x+2} = 3 + 1 + \frac{1}{3} + 0 - \frac{1}{5} - \frac{1}{3}$$
$$4 - \frac{1}{5} = \boxed{\frac{19}{5}}$$

9.

$$\sum_{b=3}^6 \frac{b^2}{(-1)^b} = -9 + 16 - 25 + 36$$
$$= \boxed{18}$$

3.

$$\left(\sum_{x=3}^7 x^2 + 2 \right) - \left(\sum_{x=3}^7 x(x-1) \right) = \sum_{x=3}^7 x^2 + 2 - (x(x-1))$$
$$\sum_{x=3}^7 x^2 + 2 - (x(x-1)) = \sum_{x=3}^7 x^2 + 2 - (x^2 - x)$$
$$\sum_{x=3}^7 x^2 + 2 - x^2 + x = \sum_{x=3}^7 x + 2$$
$$5 + 6 + 7 + 8 + 9 = \boxed{35}$$

4.

$$\begin{aligned} \left(\sum_{k=1}^5 \frac{k}{2} \right) + \left(\sum_{k=1}^5 k \right) &= \sum_{k=1}^5 -u + \frac{1}{2} - (-u^2 - u) \\ \sum_{k=1}^5 -u + \frac{1}{2} + u^2 + u &= \sum_{k=1}^5 u^2 + \frac{1}{2} \\ \frac{3}{2} + \frac{9}{2} + \frac{19}{2} + \frac{33}{2} + \frac{51}{2} &= \boxed{\frac{115}{2}} \end{aligned}$$

§A.3.3 Sections 1.6.2 and 1.7.2

1.6.2

8.

$$\begin{aligned} a_{35} &= 55 + (35 - 1)(11) \\ a_{35} &= 55 + 374 \\ a_{35} &= \boxed{429} \end{aligned}$$

10.

$$\begin{aligned} 212 &= 20 + (n - 1)(12) \\ 212 &= 20 + 12n - 12 \\ 212 &= 8 + 12n \\ 204 &= 12n \\ n &= \boxed{17} \end{aligned}$$

3.

$$\begin{aligned} a_{14} &= \left(\frac{13}{3} \right) (2^{14-1}) \\ a_{14} &= \left(\frac{13}{3} \right) (2^{13}) \\ a_{14} &= \frac{13 \cdot 8192}{3} \\ a_{14} &= \boxed{\frac{106496}{3}} \end{aligned}$$

2.

$$a_6 = 7520 + (6 - 1)(8000)$$

$$a_6 = 7520 + 40000$$

$$a_6 = \boxed{47520}$$

$$55000 - 47520 = \boxed{7480 \text{ seats were reduced.}}$$

1.7.2

1.

$$S_{16} = \frac{16}{2}(2(16.5) + (16 - 1)(6.5))$$

$$S_{16} = 8(33 + (15)(6.5))$$

$$S_{16} = 8(33 + 97.5)$$

$$S_{16} = 8(130.5)$$

$$S_{16} = \boxed{1044}$$

8.

$$\sum_{n=1}^8 \frac{1}{4^{n+4}} = \frac{1}{1024} + \frac{1}{4096} + \frac{1}{16384} + \cdots + \frac{1}{16777216}$$

$$\frac{1}{1024} \left(\frac{1 - (\frac{1}{4})^8}{1 - \frac{1}{4}} \right) = \frac{1}{1024} \left(\frac{\frac{65535}{65536}}{\frac{3}{4}} \right)$$

$$\frac{1}{1024} \cdot \frac{262140}{196608} = \boxed{\frac{262140}{201326592}}$$